

# **(TR-104) PYTHON WITH AI/ML**

## **Training Day 30 Report**

**Date: 20 February 2026**

### **Topic Covered: Production-Ready NLP System & FastAPI Model Deployment**

The thirtieth day of training focused on converting the NLP resume classifier into a production-ready industry-style project and deploying the trained machine learning model as a REST API using FastAPI.

- Converted notebook-based NLP project into a clean and modular industry-style project structure.
- Organized project into separate modules for preprocessing, feature engineering, training, prediction, models, logs, and data handling.
- Moved preprocessing logic into preprocess.py for better code reusability.
- Moved model training workflow into train.py.
- Moved prediction workflow into predict.py.
- Implemented structured logging instead of print statements for debugging and monitoring.
- Used the joblib library for saving and loading trained models and TF-IDF vectorizers.
- Built reusable prediction pipeline using predict\_resume\_category(resume\_text).
- Created a clean and structured README.md file describing project workflow and execution steps.
- Strengthened understanding of industry-standard NLP project architecture.
- Learned when to use Word Embeddings and Sentence Embeddings in NLP systems.
- Understood the role of APIs in machine learning deployment workflows.
- Built a REST API for the Resume Classifier using FastAPI.
- Created FastAPI application using app.py.
- Loaded saved classifier model and TF-IDF vectorizer inside the API service.
- Implemented home endpoint (/) and prediction endpoint (/predict).

- Accepted resume text as input and returned predicted category as JSON response.
- Tested API endpoints using Swagger UI documentation interface.
- Learned FastAPI endpoint handling and model serving concepts used in production systems.
- **GitHub Link:**  
[https://github.com/JaspinderKaurWalia26/resume\\_classifier](https://github.com/JaspinderKaurWalia26/resume_classifier)
- **Blog Link:**  
<https://medium.com/@jaspinderkaurwalia855/from-ml-model-to-rest-api-serving-a-resume-classifier-with-fastapi-734a6fd886f6>